

AI Foundation Student Handbook

A complete revision companion for the eight-domain AI Foundation course — for military personnel and civilians.

Eight domains · 43 sub-areas · 120-question final examination

Every figure in this handbook is generated from the same narration scripts as the course videos, so the two always agree.

Contents

Domain	Title	Weight	Exam questions
D1	Understanding AI: The Big Picture	12%	14
D2	How AI Works (Without the Maths)	14%	17
D3	The AI Landscape & Generative AI	12%	14
D4	Prompt Engineering Fundamentals	16%	19
D5	Working with AI Day to Day	12%	15
D6	Limitations, Risks & Hallucination	16%	19
D7	AI Security, Privacy & Data	10%	12
D8	Ethics, Bias, Responsible AI & Governance	8%	10

Total delivery: 164 minutes of video across eight domains. The final examination is 120 multiple-choice questions, 120 minutes, closed book, with a pass mark of 70 per cent.

Domain 1

Understanding AI: The Big Picture

Weight 12% · 14 exam questions · about 20 minutes

Accurate, vendor-neutral vocabulary and framing for artificial intelligence — what AI is, what it is not, and how the terms nest.

1. What AI actually is

Sub-area 1.1 — defining artificial intelligence.

2. The four nested terms

Sub-area 1.2 — AI, machine learning, deep learning and generative AI.

3. Narrow AI and general AI

Sub-area 1.3 — narrow versus general intelligence.

The test, applied

What you hear	The question to ask	Verdict
"It can write anything"	Is it one task, or every task?	Narrow
"It passed a professional exam"	Was it trained for that exam?	Narrow
"It reasons like a person"	Across every domain, no retraining?	Narrow
"It's become conscious"	Is there any evidence at all?	No

4. The AI already around you

Sub-area 1.4 — everyday artificial intelligence, and where it really shows up.

The same categories, everywhere

Where	What it's doing	The category
Your home	The assistant on your phone hears you	Speech recognition
Your office	Junk and phishing mail is filtered out	Classification
Your car	Traffic and fuel range are predicted	Forecasting
The workshop	A failing part is spotted early	Anomaly detection

Domain 1 in six lines

- AI finds patterns in data; it isn't told every rule
- Learning means guessing, comparing, adjusting — millions of times
- AI contains machine learning, which contains deep learning, which contains generative AI
- Narrow AI does one job superbly; general AI doesn't exist yet
- You already use narrow AI daily, from spam filters to fraud alerts
- Fluency isn't understanding — so we always check its work

Domain 2

How AI Works (Without the Maths)

Weight 14% · 17 exam questions · about 22 minutes

An accurate, non-mathematical model of how AI produces outputs and where it is unreliable.

1. Models, training and inference

Sub-area 2.1, models, training and inference.

Training and inference

	Training	Inference
When	Once, in advance	Every request
Dials	They adjust	Frozen still
Cost	Weeks of computing	Seconds, and cheap
Learns from you?	From its data	No, not at all

2. How language models generate text

Sub-area 2.2, how language models generate text.

3. Tokens and the context window

Sub-area 2.3, tokens and the context window.

Words, and the tokens behind them

You think in	The model counts	Rough guide
Words	Tokens	Three-quarters of a word
Characters	Tokens	About four characters
A page	Tokens	About five hundred tokens

4. Knowledge cut-off

Sub-area 2.4, the knowledge cut-off.

What the cut-off does and does not mean

Question	Answer
Is it a memory?	Patterns, not files
Is older data safe?	Still can be wrong
Can it know late news?	Only if supplied
How do you fix it?	Give it current facts

5. Variability and embeddings

Sub-area 2.5, variability and embeddings.

Domain 2 in six lines

- A model is millions of settings, tuned by training, then frozen
- Inference is simply using it, and it never learns from your chat

- Text is generated one token at a time, by predicting what comes next
- Tokens fill a finite context window - what falls off the desk is gone
- Every model has a knowledge cut-off; newer facts must be supplied
- Sampling causes variability; embeddings turn meaning into numbers

Domain 3

The AI Landscape & Generative AI

Weight 12% · 14 exam questions · about 20 minutes

A vendor-neutral map of AI capabilities and how to choose the right category of tool.

1. Generative AI and output types

Sub-area 3.1, Generative AI & Output Types.

2. Foundation models and multimodal AI

Sub-area 3.2, Foundation Models & Multimodal AI.

Inputs and outputs

Kind	Can accept	Can produce
Text-only	Typed words	Written words
Vision	Words and pictures	Written words
Multimodal	Words, pictures, sound	Words, pictures, sound

3. Retrieval-augmented generation

Sub-area 3.3, Retrieval-Augmented Generation (RAG).

4. AI agents

Sub-area 3.4, AI Agents.

Chatbot or agent?

	Chatbot	Agent
You give it	A question	A goal
It returns	An answer	Completed steps
It acts	No	Yes, using tools
Your role	Check the answer	Check the actions

5. Model, application and service

Sub-area 3.5, Model, Application & Service.

Who controls what

Layer	What it is	Who controls it
Model	The trained system	Its developer
Service	Access over a network	Its provider
Application	The product you use	Its vendor
Your use	Your task and your data	You

6. Selecting a tool category

Sub-area 3.6, Selecting a Tool Category.

Task to category

Your task	You need	Category
Draft some prose	Text output	Text generator
Work from a policy	A real source	Retrieval tool
Describe a photo	Pictures in	Multimodal tool
Repeat a small job	Action	Agent

Domain 3 in seven lines

- Generative AI creates text, images, audio or code
- Foundation models are trained once and reused widely
- Multimodal means many inputs and many outputs
- RAG answers from a source you can inspect
- An agent acts on a goal, not just an answer
- Model, application and service are three layers
- Choose the category before the product

Domain 4

Prompt Engineering Fundamentals

Weight 16% · 19 exam questions · about 24 minutes

The highest-leverage practical skill: constructing effective, reliable prompts.

1. Anatomy of a prompt

Sub-area 4.1, anatomy of a prompt.

2. Role and audience

Sub-area 4.2, role and audience.

3. Few-shot prompting

Sub-area 4.3, few-shot prompting.

Match the examples to the job

Approach	Examples given	Best for
Zero-shot	None	Simple, familiar tasks
One-shot	One example	Fixing tone or style
Few-shot	Two or three	Precise format or pattern

4. Structure and decomposition

Sub-area 4.4, structure and decomposition.

Match the approach to the task

Task size	Approach
One simple action	A single clear prompt
A few linked steps	A short sequence
A large document	Outline, then sections
Anything long	Check each step first

5. Iterative refinement

Sub-area 4.5, iterative refinement.

6. Common prompting faults

Sub-area 4.6, common prompting faults.

Fault and fix

Fault	Fix
Vague instruction	Name the exact task
No format	State the shape
Missing context	Supply the background
Unverified output	Check before you use it
Overloaded prompt	Split into steps

Domain 4 in six lines

- A prompt is a specification: task, context, format and constraints
- Role sets the perspective; audience sets the depth
- Few-shot: show two or three examples of the pattern
- Decompose big tasks into small, checkable steps
- Refine over turns — the first answer is only a draft
- Never trust unverified output — always verify at source

Domain 5

Working with AI Day to Day

Weight 12% · 15 exam questions · about 20 minutes

Practical, safe daily workflows — drafting, summarising, ideation and the verification habit.

1. Drafting and rewriting

Sub-area 5.1, drafting and rewriting.

2. Summarising

Sub-area 5.2, summarising.

What a summary tends to drop

In the document	Often lost	Keep it
A recommendation	Its conditions	Restore them
A figure	Its date or source	Cite both
A caveat	The whole caveat	Quote it
A duty	Who is responsible	Name them

3. Ideation, planning and research

Sub-area 5.3, ideation, planning and research assistance.

4. Appropriate use

Sub-area 5.4, appropriate use.

Where the line usually sits

The task	Sensible use	Keep it human
Draft a message	A first pass	Tone and facts
Summarise a policy	A rough shape	The caveats
Weigh a decision	Explore options	The decision
Personal data	Not at all	Entirely

5. The verification habit

Sub-area 5.5, the verification habit.

What to check, and against what

Check this	Against this
Facts and figures	The primary source
Any arithmetic	Your own calculation
Citations	The document itself
Names, dates, titles	The record

6. The personal prompt library

Sub-area 5.6, the personal prompt library.

What a good entry holds

Keep	Why
The prompt wording	So results repeat
A note on changes	So you improve it
The task it serves	So you can find it
A review date	So it stays current

Domain 5 in six lines

- Draft with the tool; supply the substance and keep your voice
- A summary drops caveats - restore them from the original
- Widen your ideas; never let it decide, and never let it be your source
- Appropriate use: keep private material out, and check the rules first
- Verify facts, figures, sums and citations before they leave
- Keep a small library of prompts you trust

Domain 6

Limitations, Risks & Hallucination

Weight 16% · 19 exam questions · about 22 minutes

Why fluent output can still be wrong, and how to reduce the risk.

1. Hallucination

Sub-area 6.1 — hallucination.

2. Stale information and fabrication

Sub-area 6.2 — stale information and fabrication.

3. Provenance and citation checking

Sub-area 6.3 — provenance and citation checking.

Checking a citation, step by step

Step	What you do	What it catches
Locate	Search the exact title	An invented source
Open	Read the original	A misquoted passage
Date	Check the publication date	Stale information
Attribute	Confirm the author	A fake or wrong author

4. Reducing the risk

Sub-area 6.4 — reducing the risk.

The measures, and what each one cuts

Measure	In practice	Reduces
Ground it	Supply the sources yourself	Invention
Quote it	Request the exact wording	Misquoting
Date it	Check the publication date	Staleness
Verify it	Open the source yourself	False sources
Own it	A named person decides	Unchecked use

5. Where verification is mandatory

Sub-area 6.5 — mandatory-verification contexts.

Where the check is mandatory

Context	Why the stakes rise	The check
A decision or record	The error becomes the record	A human source check
Safety, legal, money	Harm is real and lasting	A qualified person verifies
Anything briefed onward	The error travels with it	Verify before you pass it
Figures and quotations	Invented most fluently	Open the original source

Domain 6 in six lines

- Hallucination is fluent, confident and false
- Stale information is out of date, not invented
- Fabrication, including invented citations, was never true
- Check provenance: locate, open, date, attribute
- Ground it, quote it, date it, verify it, own it
- High-stakes work always needs a human source check

Domain 7

AI Security, Privacy & Data

Weight 10% · 12 exam questions · about 18 minutes

Protecting information when AI services are in the loop — data handling, shadow AI, and AI-enabled threats.

1. Protecting information

Sub-area 7.1, protecting information.

The protective ladder

Level	What it means	Public tool?
Unmarked	Never been assessed	Ask, do not assume
Official	Routine business	Often still no
Official sensitive	Would cause harm	Never
Higher still	Serious damage	Never, ever

2. Data handling by AI services

Sub-area 7.2, data handling by AI services.

Who can touch it

Stage	What happens	Under your control?
You type it in	Input captured	Yes, until you send
The service	Stored or logged	Partly, via settings
Training	Feeds the model	Only if offered
Third parties	Process or review	Rarely at all

3. Shadow AI

Sub-area 7.3, shadow AI.

4. AI-enabled threats

Sub-area 7.4, AI-enabled threats.

Signs, and the checks that still work

Threat	Warning sign	Your check
Deepfake video	Odd lighting or blinks	Call the person back
Cloned voice	Urgency, strange request	Verify on a second channel
Phishing email	Familiar tone, new ask	Check the real address
Fake invoice	Pressure to pay now	Confirm with the supplier

5. Prompt injection

Sub-area 7.5, prompt injection.

Direct and indirect

Type	Where it comes from	What happens
Direct	The user	Someone types the trick themselves
Indirect	Content the model reads	Hidden instructions are obeyed
Both	Any untrusted text	The model cannot tell them apart

Domain 7 in six lines

- Never enter protected, personal or operational data
- Assume inputs may be kept, learned from or reviewed
- Shadow AI leaks quietly; use the approved route
- Deepfakes and cloned voices are beaten out of band
- Prompt injection hides instructions inside content
- Verify anything a model says after outside reading

Domain 8

Ethics, Bias, Responsible AI & Governance

Weight 8% · 10 exam questions · about 18 minutes

Fairness, transparency, human oversight and the professional decision framework.

1. Bias and fairness

Sub-area 8.1, bias and fairness.

Four doors bias walks through

Source	What happens	Example
Data	The past is copied forward	Old hiring patterns
Coverage	Some groups are absent	A rare dialect, unlearned
Design	A proxy carries the bias	Postcode for ethnicity
Use	Applied out of context	Model moved to a new job

2. Transparency, accountability and explainability

Sub-area 8.2, transparency, accountability and explainability.

Telling the three apart

Duty	The question it answers	The simple test
Transparency	Was AI used here?	Would they know?
Explainability	Why this answer?	Can you say why?
Accountability	Who is responsible?	Is a name attached?

3. Human oversight

Sub-area 8.3, human oversight.

4. Disclosure

Sub-area 8.4, disclosure.

5. Policy and governance

Sub-area 8.5, policy and governance.

Three layers of rule

Layer	What it does	Example
Law	Sets the hard boundaries	Data protection law
Policy	Directs acceptable use	Approved tools only
Procedure	Shows how to comply	Record and review

6. A professional decision framework

Sub-area 8.6, the professional decision framework.

The framework, on one card

Question	Ask yourself	Red flag
Purpose	Is AI the right tool here?	Used just because
Permission	Does policy allow this use?	Unapproved tool
Protection	Is this data safe to share?	Sensitive data in
Proof	Have I verified the output?	Accepted unread
Fairness	Could this disadvantage anyone?	A group excluded
Responsibility	Can I explain and own it?	No name attached

Domain 8 in seven lines

- Bias enters through data, coverage, design and use
- Accurate is not the same as fair
- Transparency, explainability and accountability are three duties
- A human owns every consequential decision
- Disclose AI use whenever people are affected
- Governance: a named owner, a register, and review
- Ask: purpose, permission, protection, proof, fairness, responsibility

Glossary

Term	Meaning
Artificial intelligence	Machines performing tasks we associate with human intelligence; in practice, learning patterns from data.
Machine learning	The part of AI that improves from data rather than following rules a person wrote.
Deep learning	Machine learning using layered neural networks.
Generative AI	Deep learning that produces new content: text, images, audio or code.
Narrow AI	A system built for one task or a narrow family of tasks. All deployed AI today is narrow.
General AI	A hypothetical system matching a person across any task. It does not exist today.
Token	A chunk of text, roughly part of a word, that a language model processes.
Context window	The amount of text a model can consider at once.
Knowledge cut-off	The date after which a model has no built-in knowledge.
Embedding	A numeric representation of meaning that lets a model compare items by similarity.
Hallucination	Fluent, confident output that is simply wrong.
Prompt	The instruction you give a model.
Few-shot prompting	Including examples in a prompt to show the model the pattern you want.
Retrieval-augmented generation	Supplying a model with retrieved documents so its answer is grounded in real sources.
Agent	A system that plans and takes multiple steps towards a goal using tools.
Shadow AI	Using unapproved AI tools for official work.
Prompt injection	Hidden instructions inside content that trick a model into ignoring its real task.
Bias	A systematic unfairness in a model's output, usually inherited from its data or design.
Human oversight	A person remaining responsible for consequential decisions.
Disclosure	Telling people when AI has been used to produce something that affects them.

How to use this handbook — read the domain section before its video, then again afterwards, and use the key-point recaps for rapid revision. The videos and this handbook teach the same material in the same order.